

MOTIONCORE · DESIGN DECISION RECORD

MotionCore recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	MTC-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/motioncore		
Source	docs/decisions/0002-recommendations-accepted.md @ 1757f3f		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record every newly decided item, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Before that, MTC-DDR-001 had adopted the eleven TRL 2 recommendations for TRL 3 work, open for his review, and the TRL 3 review note (docs/REVIEW.md) listed further items as proposed, awaiting Amish. This record lists every item that the instruction turns into a decision, what changed in this repo because of it, and the items that remain open because no recommendation was made. The portfolio stays at TRL 3; TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (both sessions of 2026-09-25), MTC-DDR-001 and MTC-CAL-001. Where a recommendation named one of several options, that option is the decision.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item (source)	Decision	What changed in the repo
1 to 11	TRL 2 review items: budget scope, 20 V lower bound, VESC-class controller, DC contactor, stop category 0, interface v0.1, SwapCell heartbeat, CellGuard link, 350 W heavy case, separate firmware, CargoMule as first host (MTC-DDR-001 Table 1)	Decided as recommended; they were already built into the TRL 3 design	Wording only: "adopted for TRL 3, open for review" replaced by "decided by Amish" in MTC-DDR-001 v0.2, MTC-PRB-001 v0.4, MTC-PRC-001 v0.4, MTC-REQ-001 v0.4 and <code>bom/bom-notes.md</code> . <code>budget_usd</code> stays \$300 (item 1 redefined what the budget covers; no new figure was recommended)
15	R12 mass shortfall (MTC-DDR-001 item 15)	Option (a): relax R12 from 1.5 kg to 2.0 kg, because the enclosure is the heat sink and R9 is already at risk	MTC-REQ-001 v0.4 R12 target 2.0 kg; R12 status from not met to met (1.94 kg, 0.06 kg margin); MTC-CAL-001 v0.2 and <code>sizing.py</code> ; drawing note on MTC-DWG-001 Rev P2; key figure on concept sheet MTC-DWG-010 Rev P3; README
16	TRL 3 engineering proposals from MTC-CAL-001 (MTC-DDR-001 item 16)	Confirmed: two CAN buses on the supervisor, precharge closure at 0.6 s with a 90 % bus check, DC-rated fuses with 1 kA breaking capacity, 4 mm ² pack leads for 24 V hosts, 24 V Zener coil suppressor, controller overvoltage fault	Already in <code>bom/bom.csv</code> and MTC-PRC-001; now listed as decided design rules (MTC-PRC-001 v0.4 item 11). Because of item 17, the fuse rating moves from 58 V DC to 60 V DC or more and the overvoltage fault from 60 V to 66 V
17	R1 upper bound for CellGuard 16S LiFePO ₄ packs, 58.4 V full (cross-repo note, MTC-DDR-001 item 17)	Raise R1's upper bound from 58 V to 60 V	MTC-REQ-001 v0.4 R1 20 to 60 V; MTC-CAL-001 v0.2: overvoltage fault 66 V, bus peak after a regenerating contactor opening 62.0 V to 67.8 V (inside 75 V), direct-drive back-EMF limit 1.29 to 1.25 times no-load speed, precharge peak 0.60 A at 60 V; <code>bom/bom.csv</code> items 2 and 5; MTC-PRB-001 v0.4; MTC-PRC-001 v0.4; drawing and concept sheet notes; README
18	R10 sealing (MTC-DDR-001 item 18)	Evaluate a sealed power connector before interface v0.1 is frozen	The XT90 is marked provisional in MTC-PRC-001 v0.4 Table 2, <code>bom/bom.csv</code> item 8, MTC-DWG-001 Rev P2 and MTC-CAL-001 v0.2. The geometry is unchanged until a connector is chosen. R10 stays at risk. The evaluation itself is a paper task for a later TRL 3 session

Consequence of combining items 16 and 17. Item 16 confirmed a 60 V controller overvoltage fault, and item 17 raised the supply range to 60 V. Both cannot hold: a full CellGuard 16S pack (58.4 V) would sit 1.6 V below the fault. The overvoltage fault therefore moves to 66 V, which keeps the bus under 75 V after a contactor opening during regeneration (67.8 V), and the fuse rating follows R1 to 60 V DC. These two values are engineering consequences of Amish's decisions, recorded here so he can see them.

No design change alters geometry, mass or price: `cad/src/model.py` is unchanged and the STEP and STL files were re-exported only to confirm the model runs. The MotionCore kit stays at \$265 against \$300, and `trl` and `trl_target` stay at 3.

Items still open

Table 2. Items that remain proposed, awaiting Amish.

#	Item	Why it stays open
12	Dual-motor hosts (PalletPilot): two controllers on one supervisor, or two modules	No recommendation was made
13	Brushed motors (StepClimber): VESC DC mode or a different power stage	No recommendation was made
19	Contact coil economizer (heavy case about 55.7 °C instead of 58.5 °C)	A suggestion only in MTC-CAL-001, not a recommendation

Item 14 (whether SunSpoke keeps its own SwapCell charge-host adapter) belongs to the SunSpoke project and is listed under cross-repo actions in `docs/REVIEW.md`.

Decided but on hold (TRL 4)

Testing of the safety functions, a bench build, the timed fit for R11 and any vibration test remain TRL 4 work and are on hold by Amish's instruction. No item above required TRL 4 work to be recorded.

■ Consequences

- Requirement status on paper: ten of fourteen met, none not met, three at risk (R2, R9, R10), one not verifiable at TRL 3 (R11).
- The prospective short-circuit current of a CellGuard 16S pack is not known in this repo; it must be checked against the 1 kA fuse breaking capacity before a CellGuard host is fitted.
- CellGuard, SunSpoke and GravitySort need follow-up in their own repos (cross-repo actions in `docs/REVIEW.md`). No other repo was edited.
- Record any later change to these items as a new decision record.